Disease	Class	Ancestry	GWAS cite key	N cases	N controls	CD4+ eQTL studies	Coloc genes (PP.H4 > 0.8)
Autoimmune thyroid disease	IMD	EUR	saevarsdottir2020	30,234	725,172	7	63
Crohn's disease	IMD	EUR	delange2017	12,194	28,072	7	28
Childhood-onset asthma	IMD	EUR	ferreira2019	13,962	300,671	7	27
Type 1 diabetes	IMD	Multi	robertson2021	22,153	37,374	7	26
Rheumatoid arthritis (EUR)	IMD	EUR	ishigaki2022	22,350	74,823	7	25
Inflammatory bowel disease	IMD	EUR	delange2017	25,042	34,915	5	24
Adult-onset asthma	IMD	EUR	ferreira2019	26,582	300,671	5	23
Atopic dermatitis	IMD	EUR	buduaggrey2023	60,653	804,329	7	18
Rheumatoid arthritis (EAS)	IMD	EAS	ishigaki2022	11,025	162,608	7	16
Asthma (EAS)	IMD	EAS	sakaue2021	13,015	162,933	6	9
Ulcerative colitis	IMD	EUR	delange2017	12,366	33,609	6	9
Systemic lupus erythematosus	IMD	EUR	bentham2015	5,201	9,066	5	7
Type 2 diabetes	non-IMD	Multi	xue2018	62,892	596,424	5	21
Myocardial infarction	non-IMD	Multi	sakaue2021	35,909	587,120	5	20
Schizophrenia	non-IMD	EUR	pardinas2018	40,675	64,643	5	17
Osteoarthritis	non-IMD	Multi	mcdonald2022	133,506	335,938	5	5
Alzheimer's disease (proxy)	non-IMD	EUR	jansen2019	71,880	383,378	2	3
Ischaemic stroke	non-IMD	Multi	sakaue2021	34,593	624,214	2	3

Table 2.1

Eighteen GWAS traits (12 immune-mediated, 6 non-IMD controls) colocalise with 2–7 CD4+ eQTL studies and contribute 3–63 coloc genes (anchors).

Complete LaTeX table float (caption + \label{tab:traits}); \input it and cite it with Table~\ref{tab:traits}. Needs booktabs and tabularx. Columns: disease, ancestry, GWAS (\cite), cases, controls, number of CD4+ T-cell eQTL studies in which the GWAS signal colocalised with an eQTL (PP.H4 > 0.8), number of coloc genes. Immune-mediated diseases first, then non-IMD controls, each by coloc genes. Note under the table: traits from the same GWAS share samples (de Lange 2017; Ferreira 2019; Ishigaki 2022). Per-disease eQTL study names are in data/table_2_1_eqtl_sources.csv (study mapping unverified).

File: main/table_2_1_traits.tex, main/table_2_1_traits.csv
Section: Traits and anchors
Command: uv run 01_coloc_summary_table.py

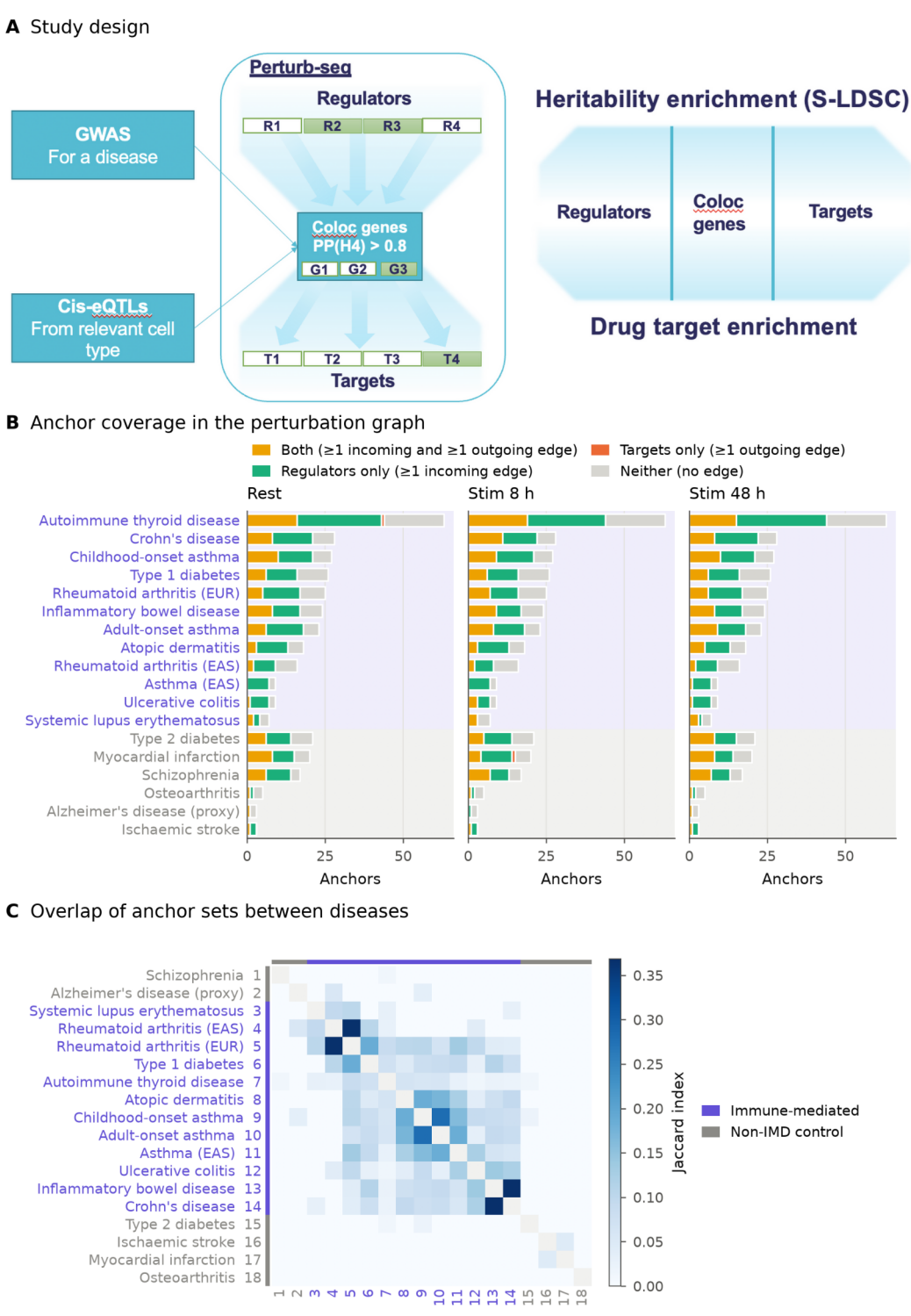


Fig. 2.1

Most anchors respond to some knockdown but few are strong knockdowns themselves, and anchor sets overlap mainly between related immune-mediated diseases.

(A) Study design: GWAS and cis-eQTL colocalisation gives coloc genes (anchors, PP.H4 > 0.8); Perturb-seq expands them to regulators (knockdowns that change an anchor) and targets (genes changed when an anchor is knocked down); each set is tested for heritability enrichment (S-LDSC) and drug-target enrichment. (B) Anchors per disease and condition, classified by their edges in the perturbation graph (|logFC| ≥ 0.5): both incoming and outgoing (yellow), incoming only (green), outgoing only (orange), or none (grey). * no targets in any condition (no anchor has an outgoing edge). (C) Jaccard index of anchor sets between diseases, ordered by average-linkage clustering; columns are numbered as the rows; diagonal masked. Disease names are coloured by class (purple, immune-mediated; grey, non-IMD control).

File: main/fig_2_1_overview.pdf
Section: Study design and anchors
Command: uv run 07_fig_2_1.py

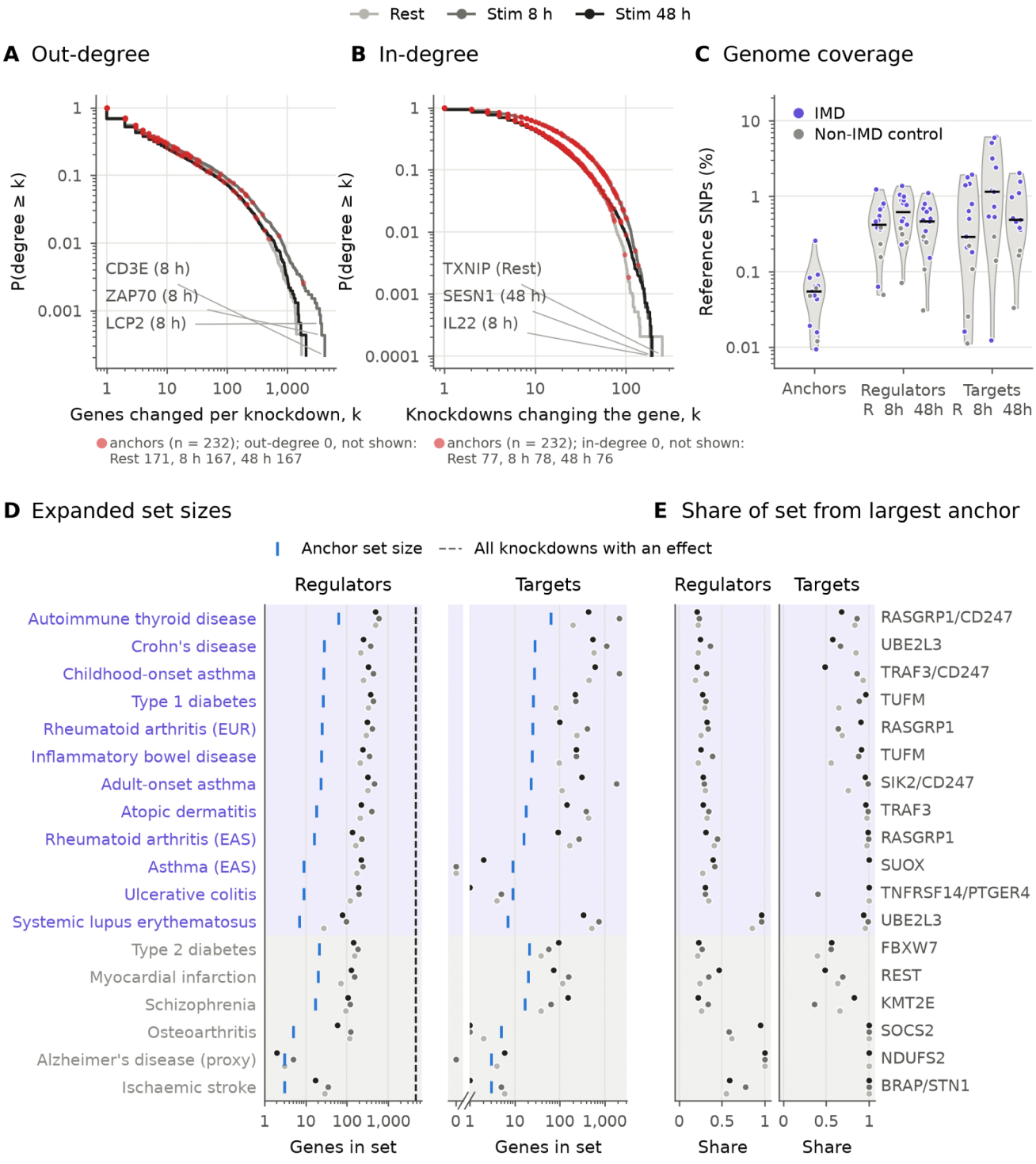


Fig. 2.2

A few knockdowns affect many genes and some genes respond to many knockdowns, so expanded sets are large and often come from a single anchor.

(A, B) Complementary CDFs of out-degree (genes changed per knockdown; n = knockdowns with an effect, overview_numbers.csv regulator_universe) and in-degree (knockdowns changing a gene); the three largest hubs are labelled with the condition in which they occur. Red points: the anchors (union of coloc genes over all traits) on their condition's curve; anchors with degree 0 cannot be drawn on the log axis and are counted below each panel. (C) Percentage of reference SNPs in the anchor, regulator and target annotations: violins (density estimated on the log scale), one point per disease, black bar at the median (fitted S-LDSC sets, incl. the single-set runs for osteoarthritis regulators and Alzheimer's proxy 48 h targets; no results for schizophrenia, the Alzheimer's proxy at Rest and 8 h, or sets under 1,000 SNPs). (D) Regulator and target set sizes per disease and condition; blue ticks, anchor set size; dashed lines, all knockdowns with an effect. Empty target sets are shown in the '0' strip left of the axis break. (E) Share of each expanded set contributed by its largest anchor; dominant anchors are named where the target share exceeds 0.5. Empty sets are omitted: Asthma (EAS), Ulcerative colitis, Osteoarthritis, Ischaemic stroke (all conditions), Alzheimer's disease (proxy) (Rest, 8 h), Type 2 diabetes (Rest).

File: main/fig_2_2_graph_and_sets.pdf
Section: Perturbation-response graph and expanded sets
Command: uv run 05_perturbation_graph.py

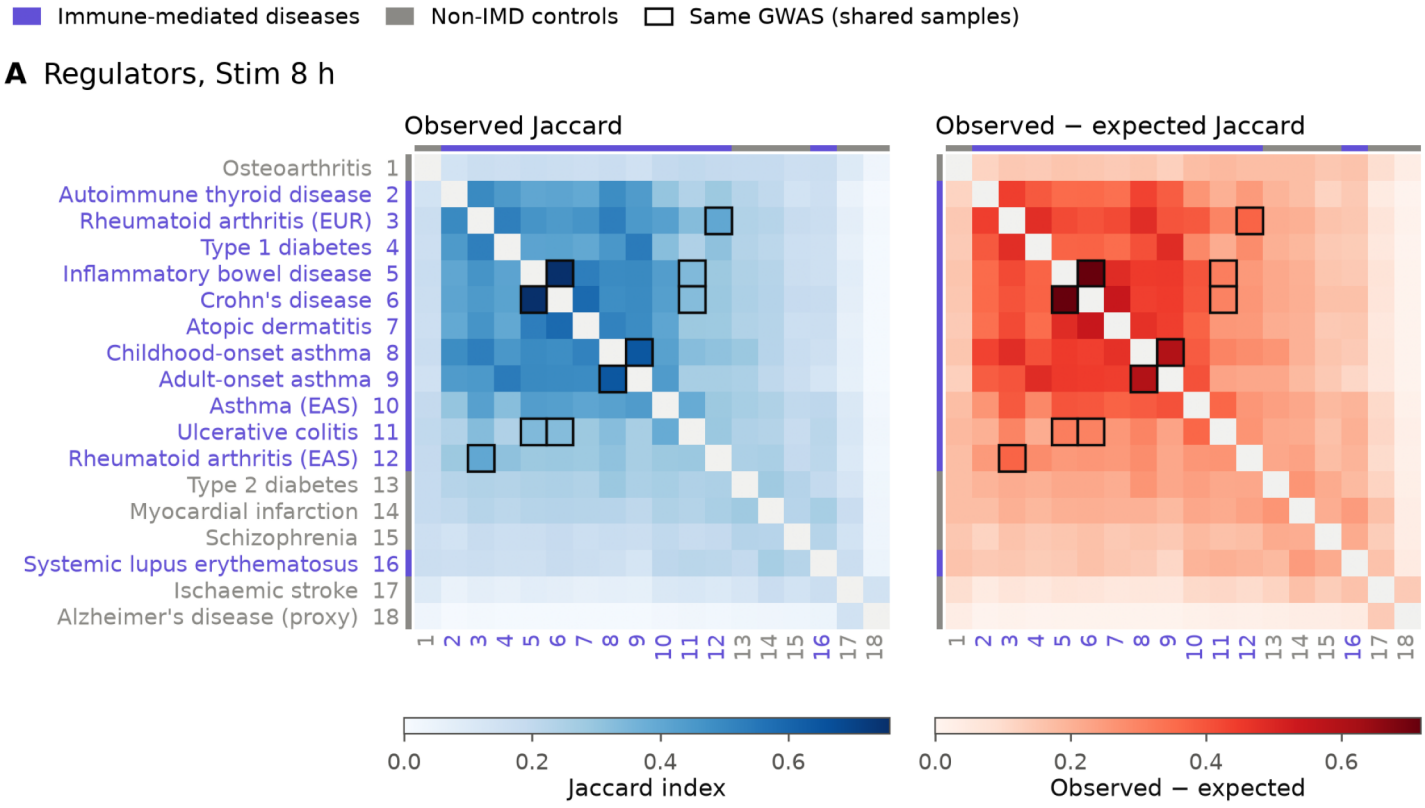


Fig. 2.6A

At 8 h, regulator sets overlap more than expected by chance for every pair of diseases, most strongly among immune-mediated diseases.

Observed Jaccard index (left) and observed minus expected Jaccard (right) between regulator sets at 8 h of stimulation. The expectation is for random sets of the same sizes drawn from all knockdowns with an effect; no pair falls below expectation at 8 h. Rows and columns share one order (clustering on mean observed Jaccard across conditions); columns are numbered as the rows; diagonal masked. Black outlines mark pairs of traits from the same GWAS, which share samples (de Lange 2017; Ferreira 2019; Ishigaki 2022).

File: main/fig_2_6a_regulator_overlap_8h.pdf
Section: Overlap of expanded sets between diseases
Command: uv run 04_jaccard_heatmaps.py --condition Stim8hr --pair

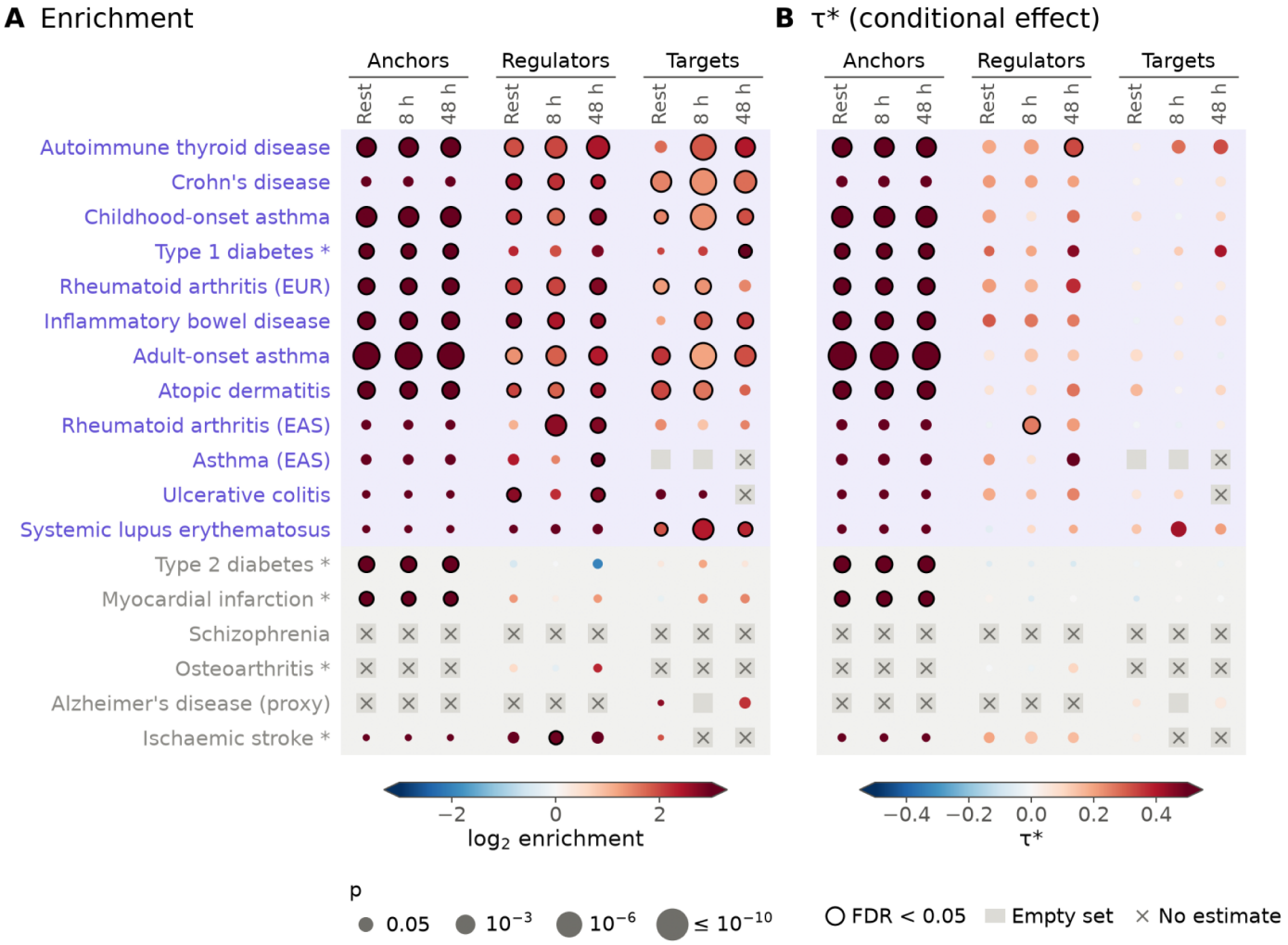


Fig. 4

Expanded sets are marginally enriched for immune-mediated disease heritability, but not conditionally on the baseline and anchor annotations.

S-LDSC for anchor, regulator and target sets (baselineLD v2.2 plus the three sets fitted jointly per trait and condition). (A) Heritability enrichment: colour, log2 enrichment (clipped at ±3); dot area, −log10 p (two-sided). (B) τ*, the standardised per-annotation effect conditional on all other annotations in the model (clipped at ±0.5); dot area, −log10 one-sided p (H1: τ* > 0). Black outline, BH FDR < 0.05, corrected within regulator + target sets (84 tests) and within anchors (45 tests) separately, for A and B. Lupus regulators at Rest have negative Prop._h2 (enrichment −11.8, drawn at the blue end) and are not counted as significant. Grey square, empty set; grey square with ×, no estimate (schizophrenia: S-LDSC failed; Alzheimer's proxy at Rest and 8 h: not run; sets under 1,000 SNPs dropped before fitting). Rows in the order of Fig. 2.2D. * multi-ancestry GWAS analysed with the European LD reference; East Asian GWAS used the East Asian reference.

File: main/heritability/fig_4_heritability.pdf
Section: Heritability enrichment of expanded sets
Command: uv run heritability/02_fig_4_heritability.py

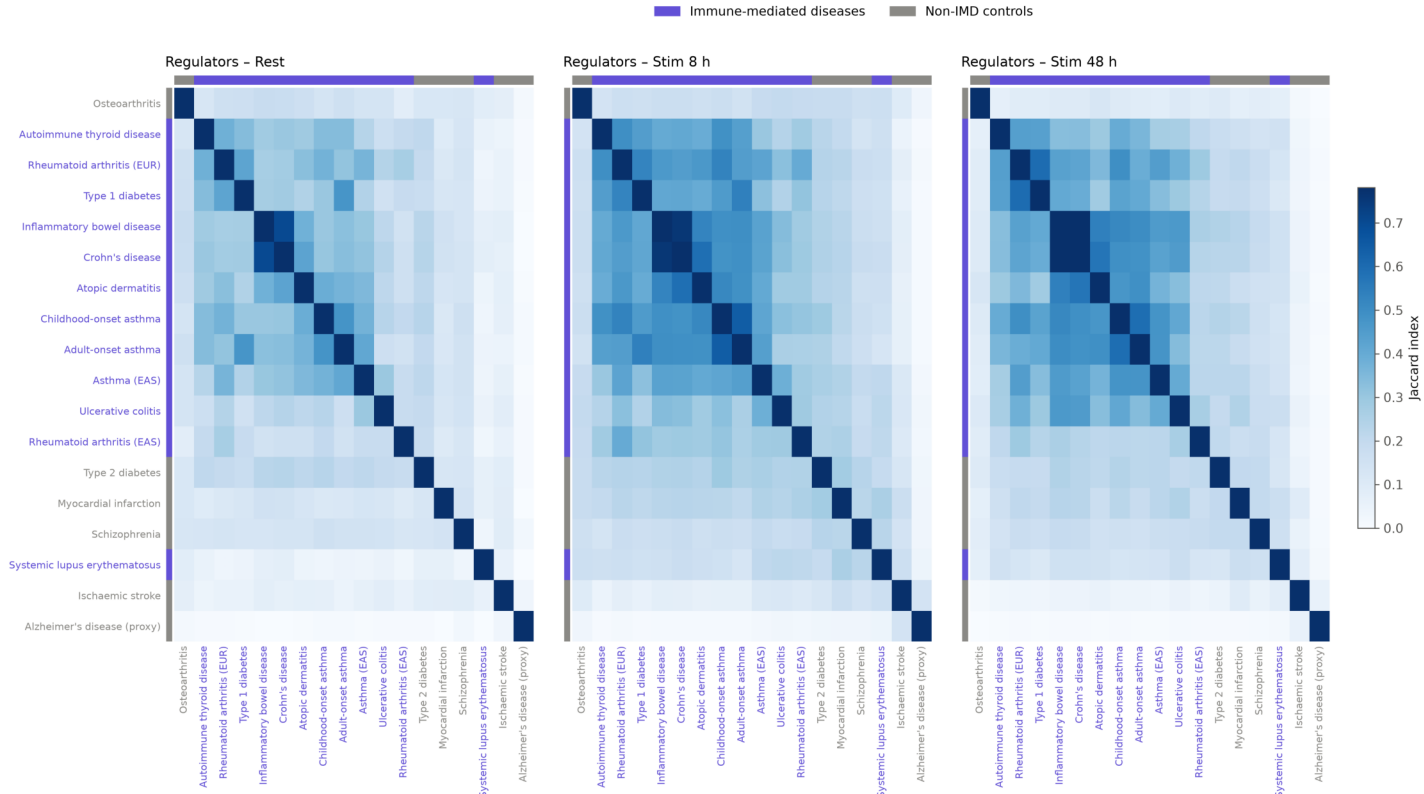


Fig. S2A

Regulator-set overlap between diseases is similar in all three conditions.

Observed Jaccard index between regulator sets at Rest, 8 h and 48 h, shared ordering.

File: supp/fig_s2a_regulator_overlap_observed.pdf
Section: Overlap of expanded sets between diseases
Command: uv run 04_jaccard_heatmaps.py

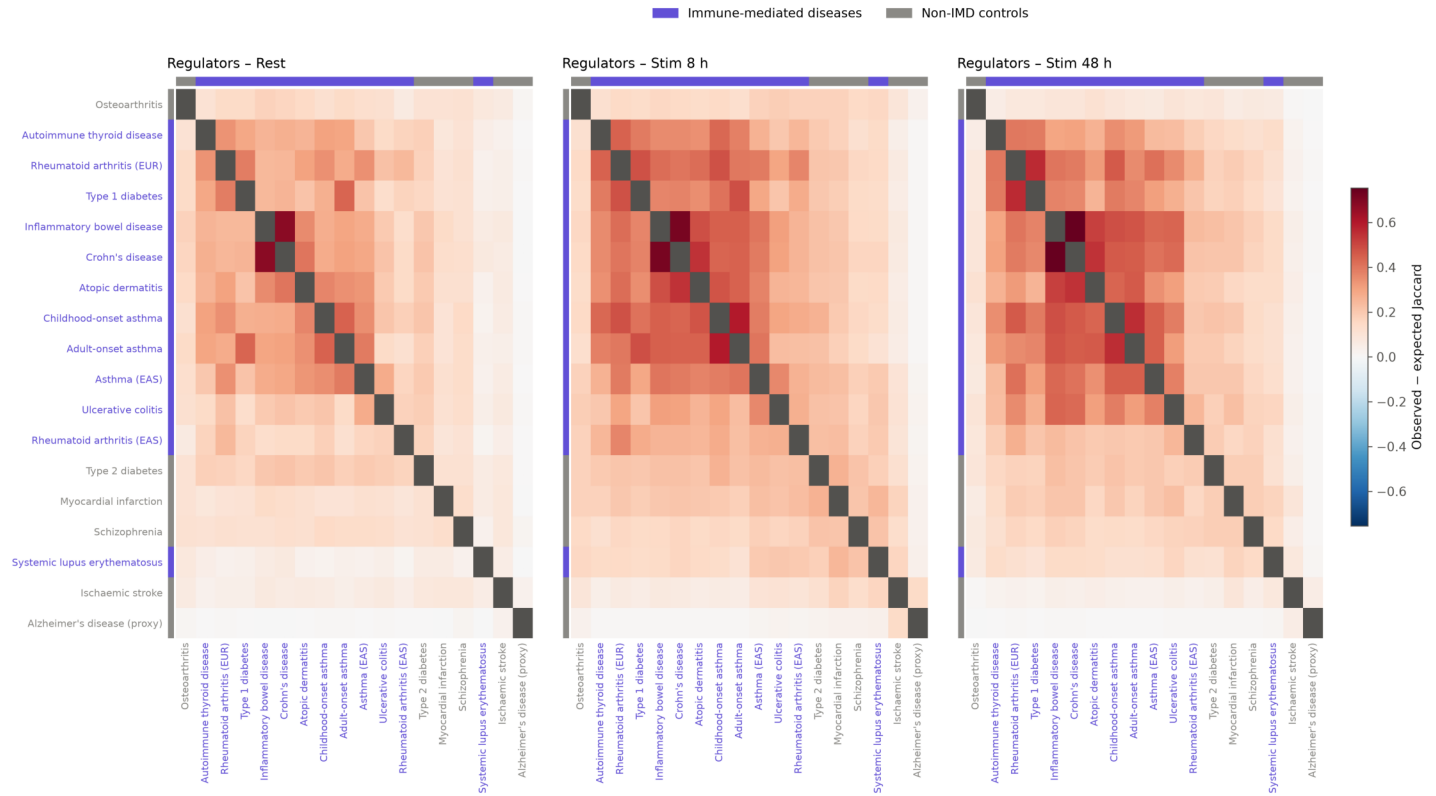


Fig. S2B

In every condition, regulator-set overlap exceeds the random expectation for almost all disease pairs, most for pairs of immune-mediated diseases.

Observed minus expected Jaccard between regulator sets per condition (universe: knockdowns with ≥1 edge in that condition).

File: supp/fig_s2b_regulator_overlap_excess.pdf
Section: Overlap of expanded sets between diseases
Command: uv run 04_jaccard_heatmaps.py --excess

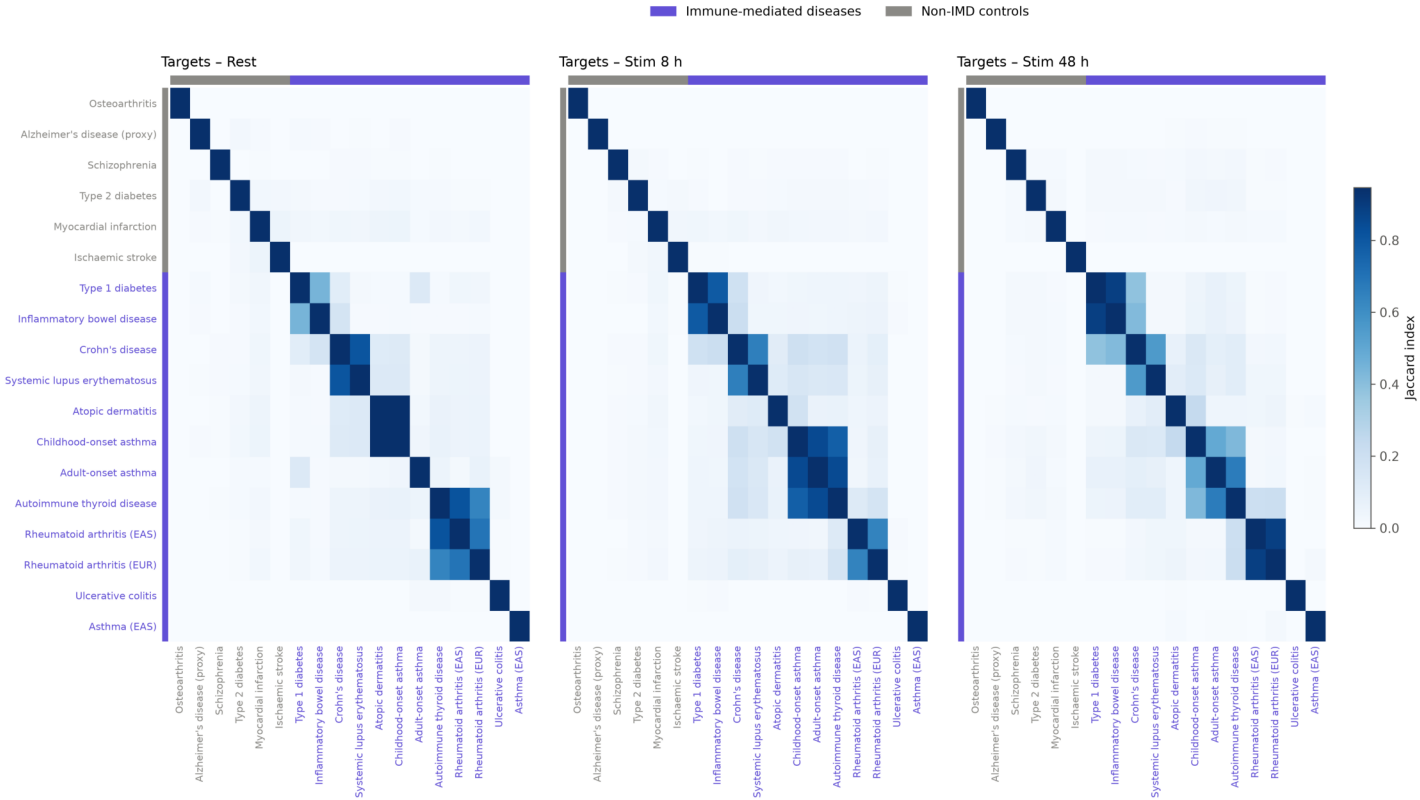


Fig. S2C

Target sets overlap only between diseases that share a dominant anchor.

Observed Jaccard index between target sets at Rest, 8 h and 48 h, shared ordering.

File: supp/fig_s2c_target_overlap_observed.pdf
Section: Overlap of expanded sets between diseases
Command: uv run 04_jaccard_heatmaps.py

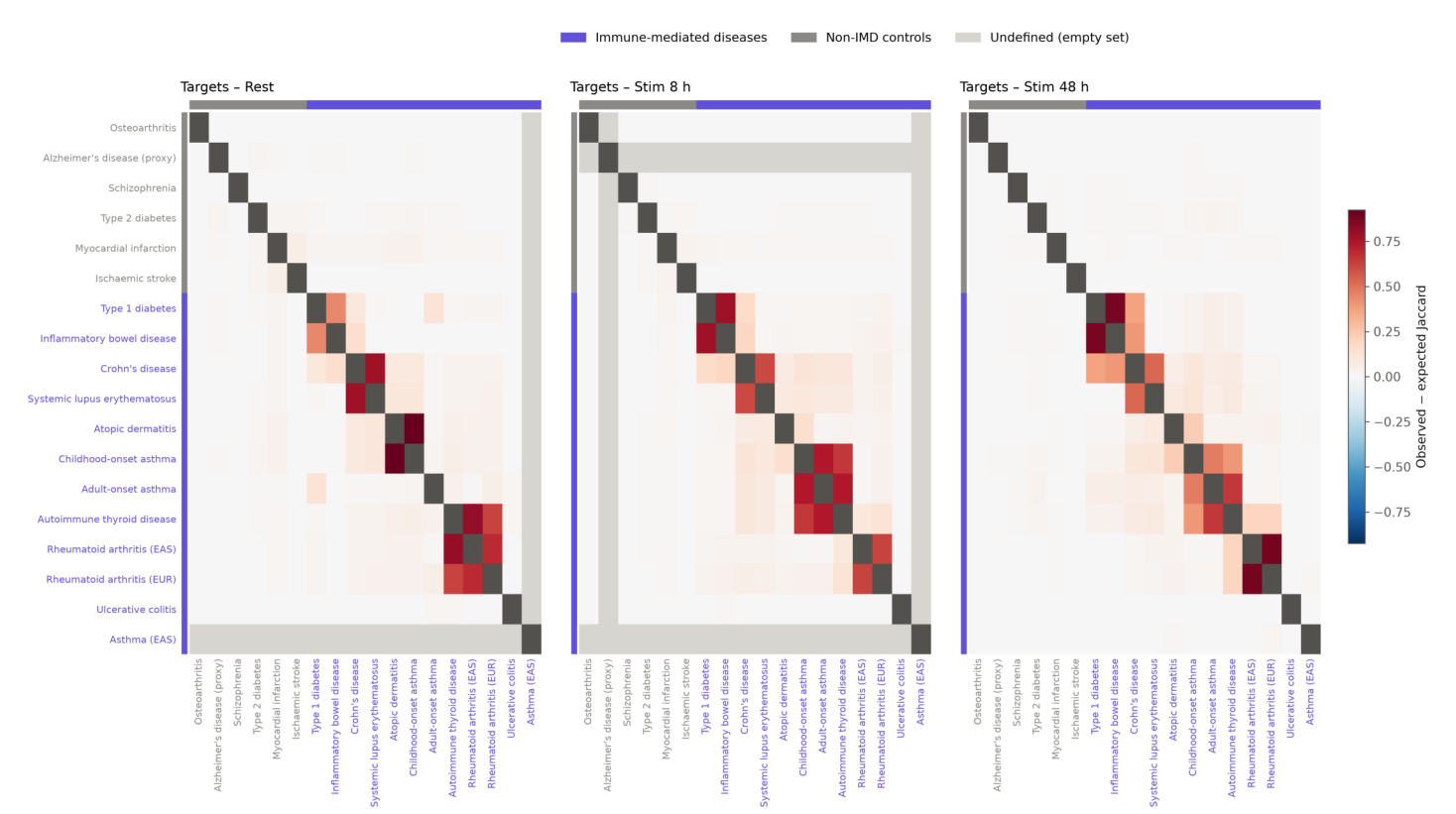
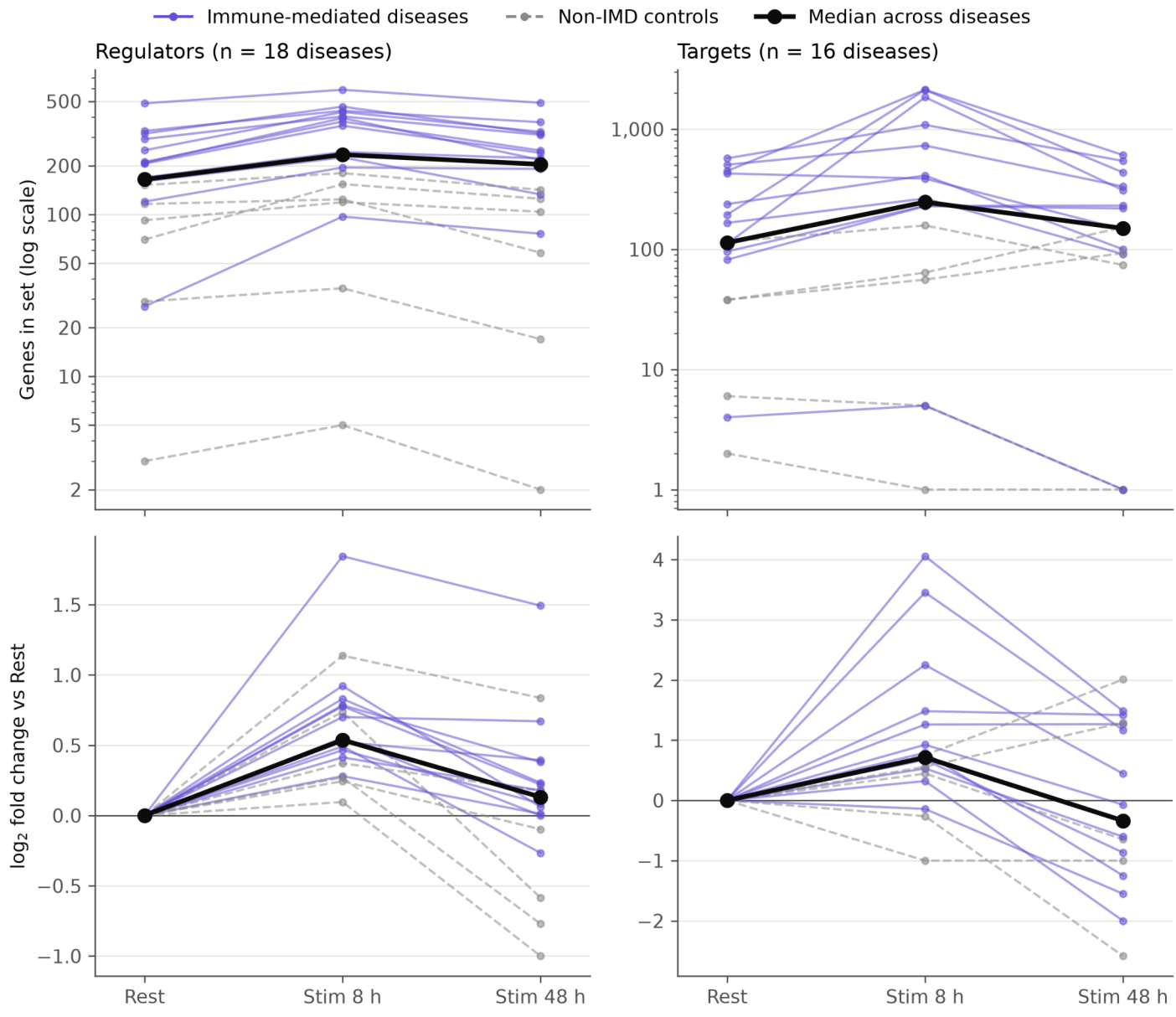


Fig. S2D

Excess target-set overlap is near zero except for diseases that share a dominant anchor.

Observed minus expected Jaccard between target sets per condition (universe: genes with ≥1 incoming edge). Grey, undefined (a disease with no targets).

File: supp/fig_s2d_target_overlap_excess.pdf
Section: Overlap of expanded sets between diseases
Command: uv run 04_jaccard_heatmaps.py --excess



Targets: excluded (empty set in ≥1 condition): Alzheimer's disease (proxy), Asthma (EAS).

Fig. S3

Regulator and target sets are largest at 8 h of stimulation.

Set size per disease across Rest, 8 h and 48 h (top, log scale) and log2 fold change versus Rest (bottom); black, median across diseases. Diseases with an empty set in any condition are excluded (listed under the figure). Friedman / Wilcoxon results are in data/set_size_by_condition_tests.csv, not on the figure (diseases are not independent).

File: supp/fig_s3_set_size_by_condition.pdf
Section: Expanded sets across stimulation conditions
Command: uv run 03_condition_size_change.py

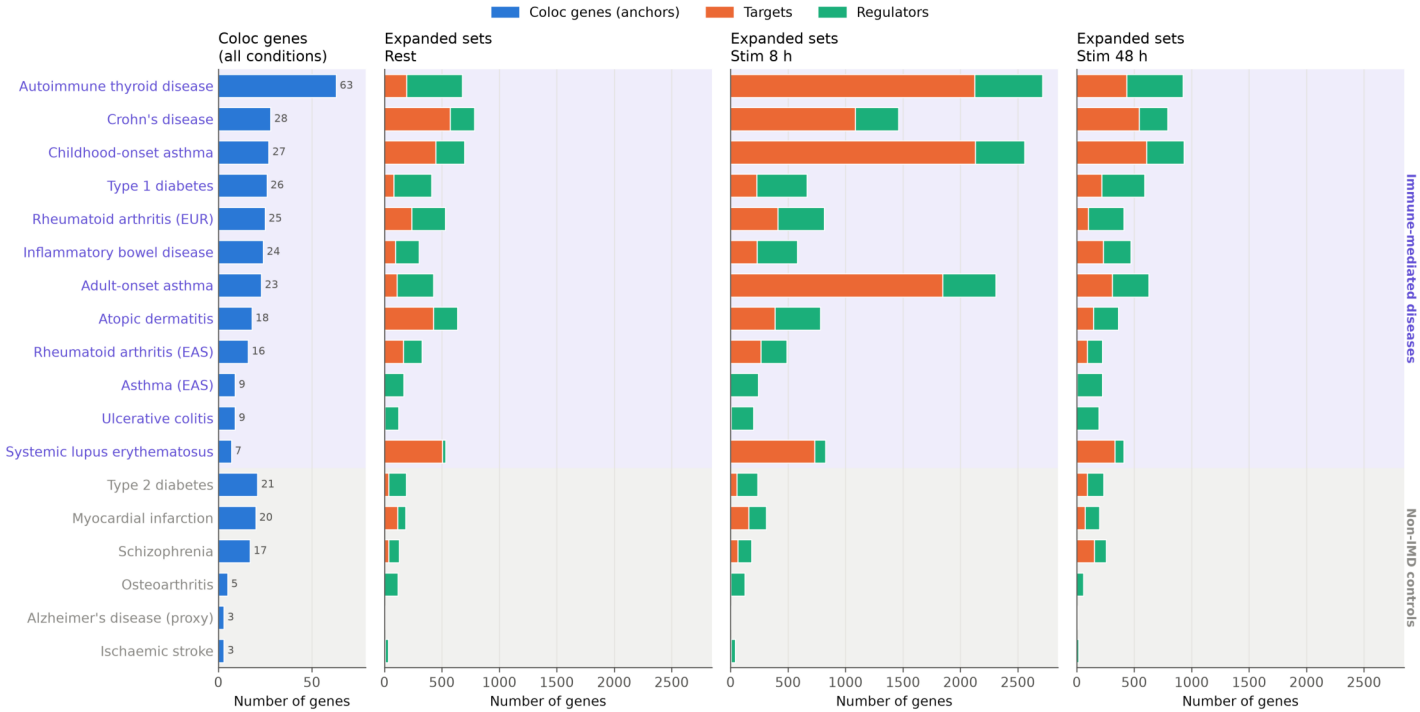


Fig. S4

Expanded sets are one to two orders of magnitude larger than anchor sets.

Coloc genes (own axis, condition-independent) and stacked target + regulator set sizes per condition. Stacked length is not the union size: regulators and targets can share genes.

File: supp/fig_s4_stacked_set_sizes.pdf
Section: Perturbation-response graph and expanded sets
Command: uv run 02_gene_set_sizes_barplot.py

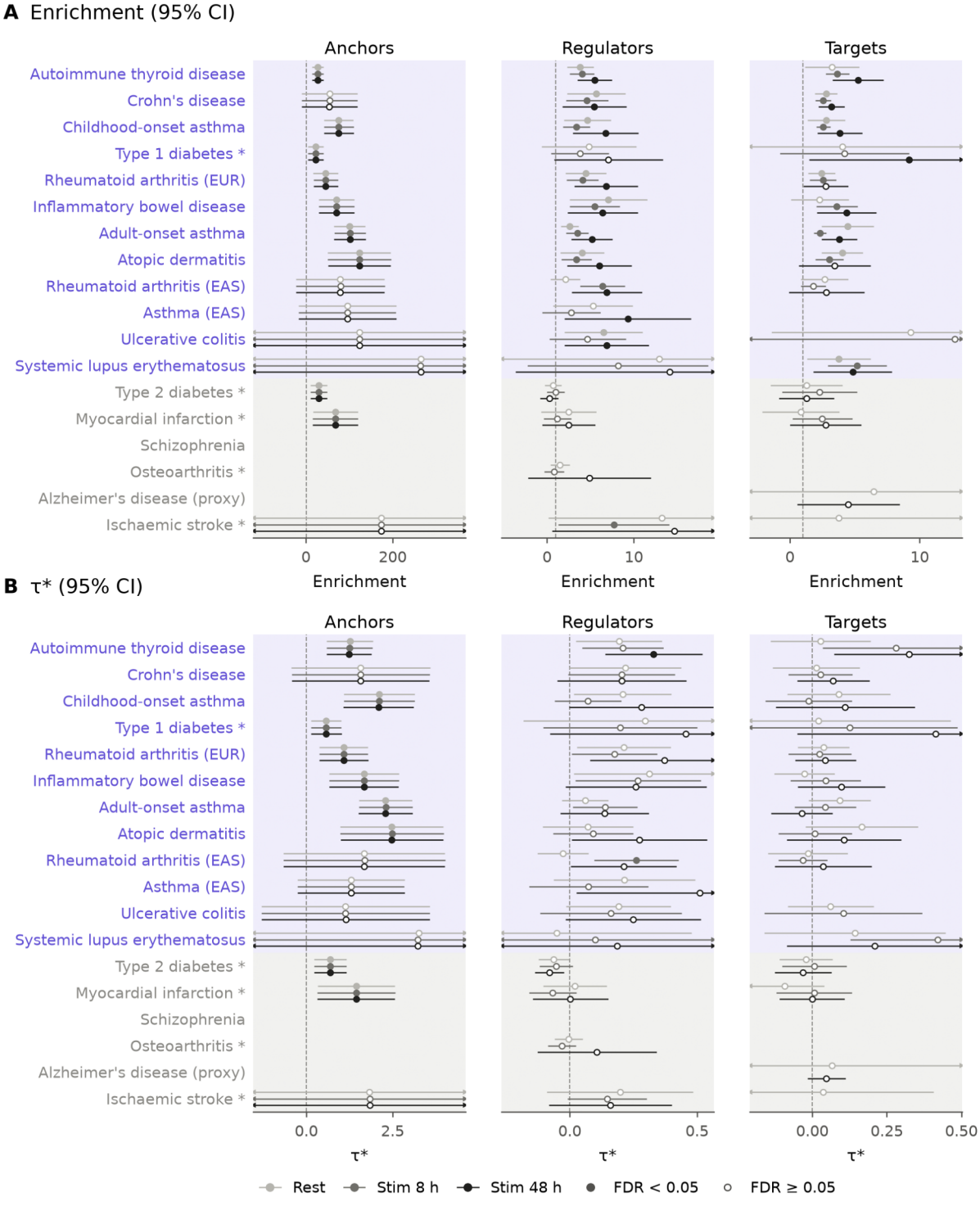


Fig. S5

Expanded-set enrichment intervals mostly exclude one, while their τ intervals mostly include zero.*

Enrichment (A) and τ* (B) with 95% CIs per trait, annotation and condition (colour); filled points significant as in Fig. 4. Each panel's x range covers the bulk of the estimates; CIs past it end in a small arrowhead at the panel edge, and estimates past it are drawn as a larger arrowhead. * as in Fig. 4.

File: supp/heritability/fig_s_heritability_forest.pdf
Section: Heritability enrichment of expanded sets
Command: uv run heritability/03_fig_s_forest.py

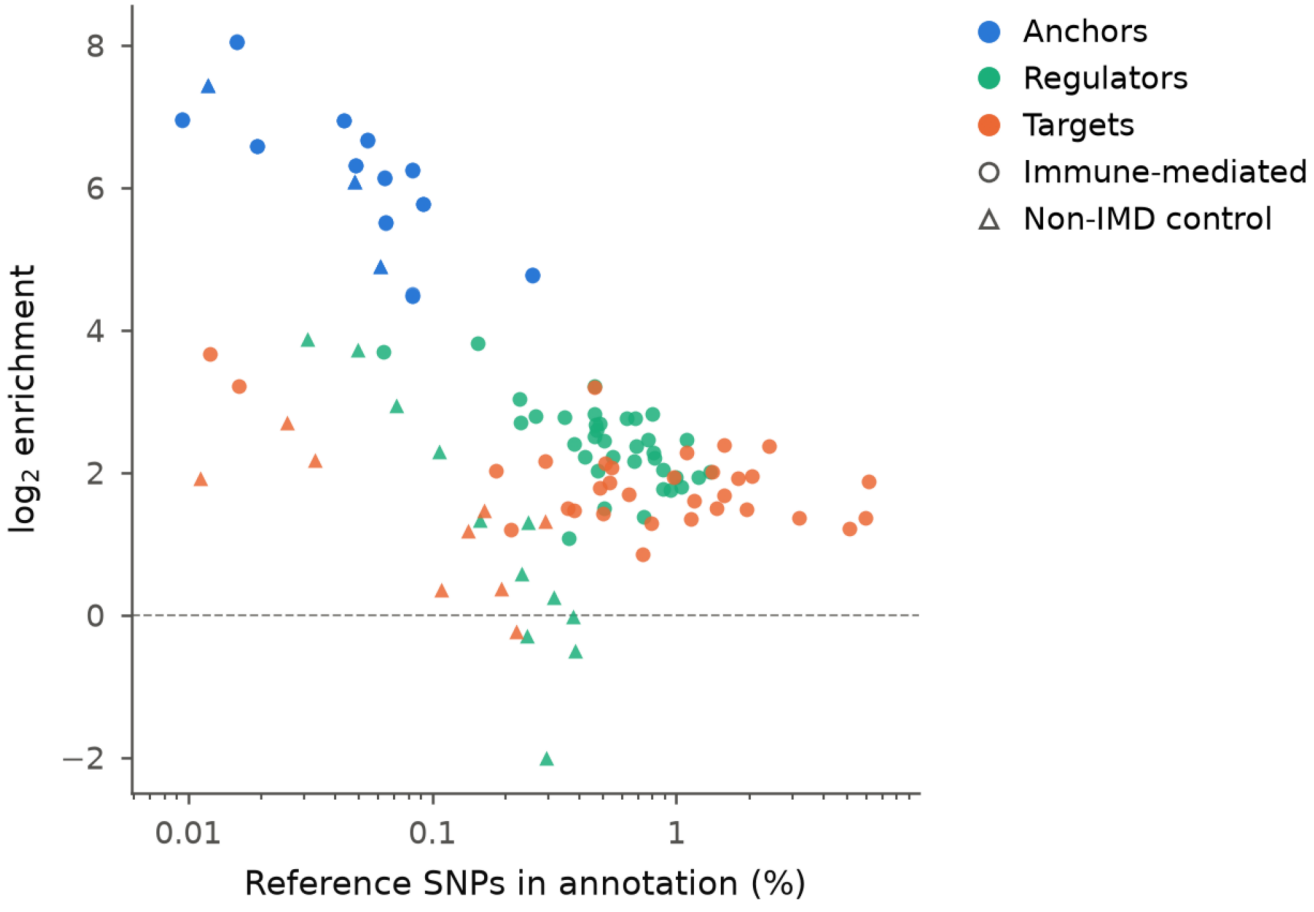


Fig. S6

Enrichment falls as annotations get larger: small anchor sets have the largest enrichment.

log2 enrichment against % of reference SNPs in each fitted set; colour, annotation; shape, disease class. Lupus regulators at Rest (enrichment −11.8) are not shown (no log2).

File: supp/heritability/fig_s_enrichment_vs_coverage.pdf
Section: Heritability enrichment of expanded sets
Command: uv run heritability/04_fig_s_scatter.py

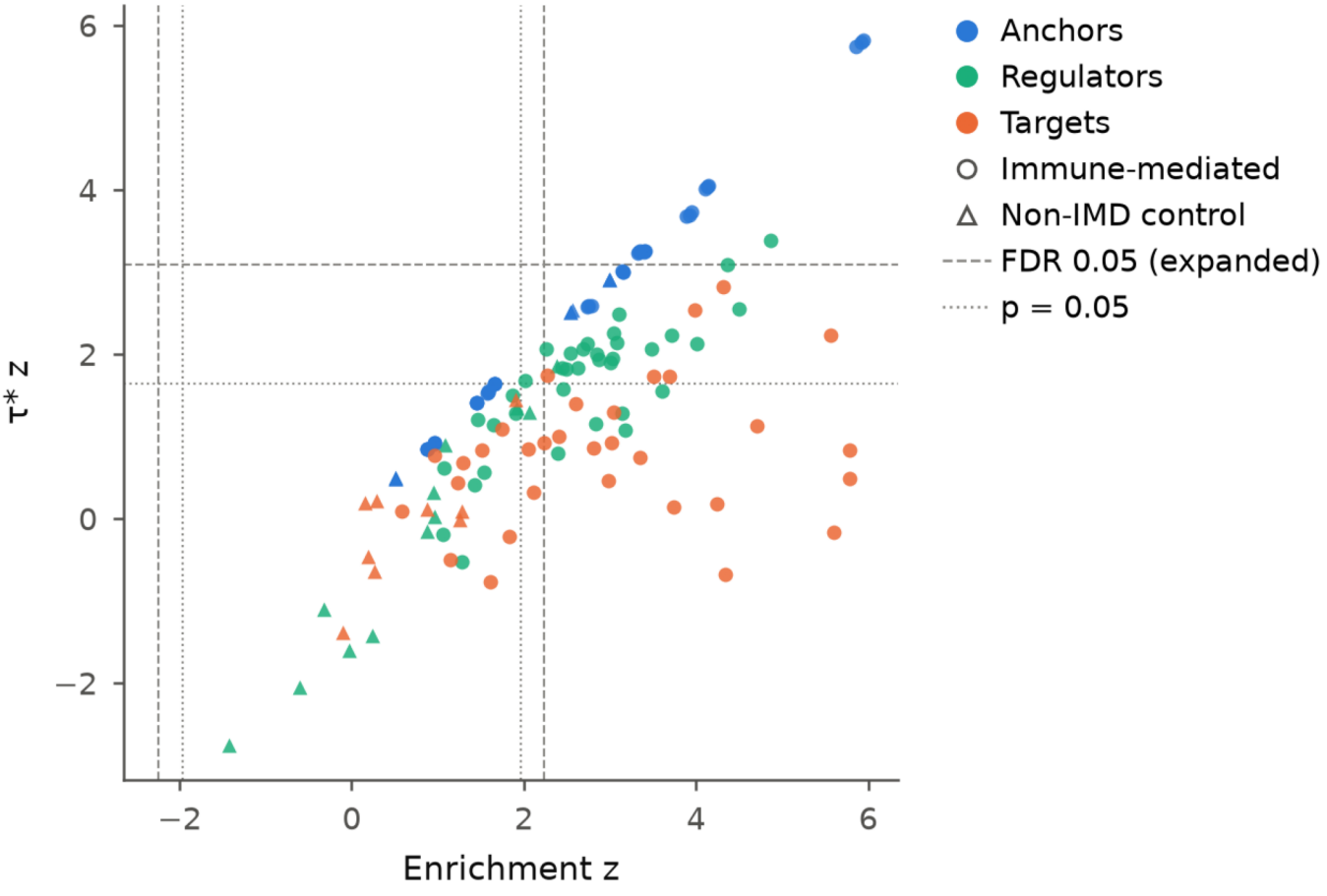


Fig. S7

No enriched regulator or target set passes the τ threshold; anchors pass both or neither.*

Signed enrichment z (from the S-LDSC enrichment p) against τ* z per fitted set. Dashed lines, FDR 5% thresholds for regulator + target sets (enrichment two-sided, |z| = 2.19; τ* one-sided, z = 3.24, the BH hurdle for the smallest p since no set passes); dotted, nominal p = 0.05 (|z| = 1.96; one-sided z = 1.64). Anchors are corrected separately (lines not drawn). Bottom-left point: lupus regulators at Rest (negative Prop._h2).

File: supp/heritability/fig_s_enrichment_vs_tau.pdf
Section: Heritability enrichment of expanded sets
Command: uv run heritability/04_fig_s_scatter.py

All files in figures/

Table 2.1	main/table_2_1_traits.tex
Table 2.1	main/table_2_1_traits.csv
Fig. 2.1	main/fig_2_1_overview.pdf
Fig. 2.2	main/fig_2_2_graph_and_sets.pdf
Fig. 2.6A	main/fig_2_6a_regulator_overlap_8h.pdf
Fig. 4	main/heritability/fig_4_heritability.pdf
Fig. S2A	supp/fig_s2a_regulator_overlap_observed.pdf
Fig. S2B	supp/fig_s2b_regulator_overlap_excess.pdf
Fig. S2C	supp/fig_s2c_target_overlap_observed.pdf
Fig. S2D	supp/fig_s2d_target_overlap_excess.pdf
Fig. S3	supp/fig_s3_set_size_by_condition.pdf
Fig. S4	supp/fig_s4_stacked_set_sizes.pdf
Fig. S5	supp/heritability/fig_s_heritability_forest.pdf
Fig. S6	supp/heritability/fig_s_enrichment_vs_coverage.pdf
Fig. S7	supp/heritability/fig_s_enrichment_vs_tau.pdf
Table 2.1	data/table_2_1_eqtl_sources.csv
Fig. 2.1B	data/anchor_coverage.csv
Fig. 2.1C	data/jaccard_anchors_observed.csv
Fig. 2.2A, B	data/degrees.csv
Fig. 2.2D, S3, S4	data/set_sizes.csv
Fig. 2.2C	data/genome_coverage.csv
Fig. 2.2E	data/largest_anchor_share.csv
Fig. 2.2 (text)	data/overview_numbers.csv
Fig. S3	data/set_size_by_condition_tests.csv
Fig. S2A, S2B	data/jaccard_regulators_Rest_observed.csv
Fig. S2A, S2B	data/jaccard_regulators_Rest_expected.csv
Fig. 2.6A, S2A, S2B	data/jaccard_regulators_Stim8hr_observed.csv
Fig. 2.6A, S2A, S2B	data/jaccard_regulators_Stim8hr_expected.csv
Fig. S2A, S2B	data/jaccard_regulators_Stim48hr_observed.csv
Fig. S2A, S2B	data/jaccard_regulators_Stim48hr_expected.csv
Fig. S2C, S2D	data/jaccard_targets_Rest_observed.csv
Fig. S2C, S2D	data/jaccard_targets_Rest_expected.csv
Fig. S2C, S2D	data/jaccard_targets_Stim8hr_observed.csv
Fig. S2C, S2D	data/jaccard_targets_Stim8hr_expected.csv
Fig. S2C, S2D	data/jaccard_targets_Stim48hr_observed.csv
Fig. S2C, S2D	data/jaccard_targets_Stim48hr_expected.csv
Fig. 4, S5–S7	data/heritability/heritability_long.csv
Fig. 4 (methods)	data/heritability/ldsc_runs.csv
Text	data/numbers_for_text.md
—	INDEX.md
—	contact_sheet.pdf
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All files in figures/ (continued)

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